ENT-164 · Intro to Making

Tufts University | Derby Entrepreneurship Center at Tufts

CLASS 9 · WEEK 9 · FALL 2026

The Final Project Sprint

The Robo Challenge is in the books. A three-hour sprint: from idea to a prototype plan. Brainstorm with your team, choose a concept, and leave with an idea for what you'll build.

Thu, Nov 5 | JCC 301 / Nolop | Concept generation | Note-and-Vote



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Tufts University · Gordon Institute

Robo Challenge awards

You designed, built, and raced working robots. Here's how it ended.

The race

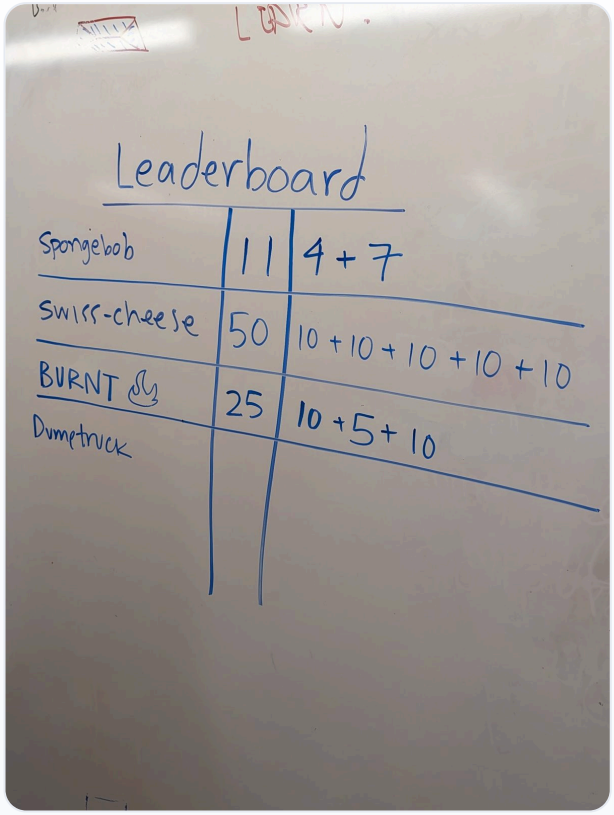
Teams moved as many balls as possible between the two lines, with robots they built from scratch.

The results

Scores are in, and the awards go out to the teams behind this semester's robots.

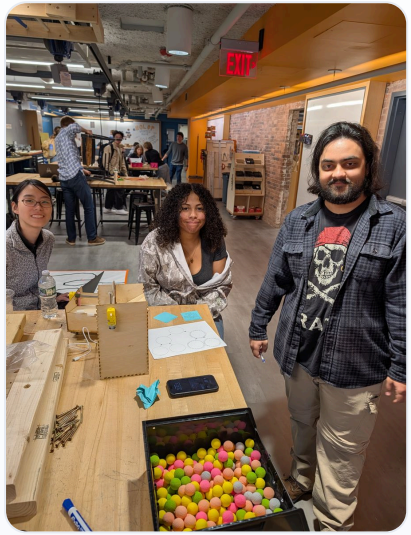
Carry it forward

Reflect on what worked, what didn't, and what you'd do differently, then bring that into the final project.



It's not about winning. It's about learning

... and having fun!

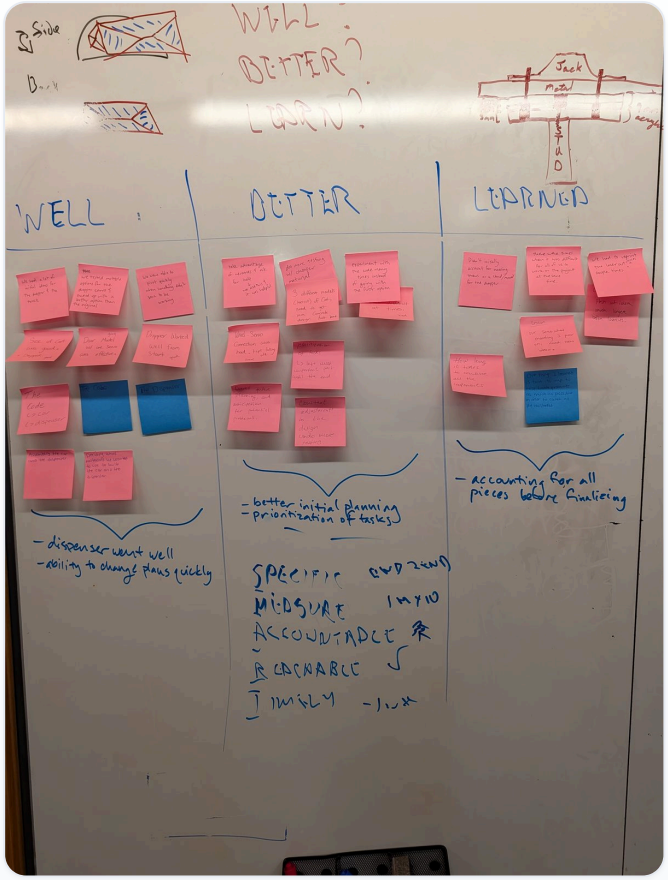


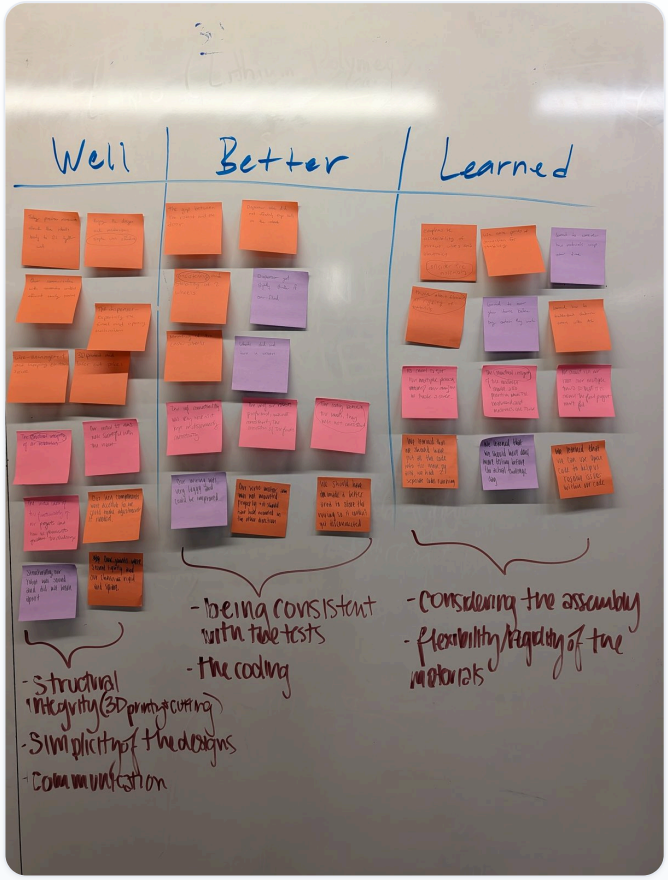






You learned a lot





Robo Challenge retro

WHAT WENT WELL

Communication and teamwork
Clear roles, frequent check-ins.

Adapting on the fly
Rubber bands for grip, spare power bank.

Early prototyping paid off
Foam models settled shape and material.

Solid fabrication
Laser-cut and printed parts fit as planned.

Simple, reused designs
One proven gate mechanism, used twice.

KEY CHALLENGES

Little full-system testing
Full runs came at the last minute.

Electronics underestimated
Power drain, wiring, WiFi, motor control.

Hard to assemble and reach
Glued batteries, hidden power switch.

Late material choices
Warped prints, flimsy single mounts.

No slack in the plan
Work finished right at the deadline.

LEARNINGS

Schedule full test runs
Two or three complete runs, a week early.

Start code and wiring early
Prototype with real motors and batteries.

Design for access
Reachable power. Temporary fixings first.

Match material to the load
Wood for structure, more mount points.

Owners and weekly check-ins
SMART goals. Riskiest part first.

Five weeks to Demo Day

What happens between today and the final showcase.

WEEK 9

The Final Project Sprint

Nov 5

Brainstorm & Note-and-Vote

→

WEEK 10

From Device to Cloud

Nov 12

Sketch, mockup & 5-minute pitch

→

WEEK 11

Final Project Making Time

Nov 19

Start making

→

BREAK

Thanksgiving

Nov 26

No class

→

WEEK 12

Final Project Making Time

Dec 3

Integrate & test runs at Nolop

→

WEEK 13

Final Project Showcase

Dec 10

Present the final product

DELIVERABLE

Concept presentation. 5 Minutes per team: problem & opportunity, proposed product concept with sketch and mockup, and next steps.

Build sessions happen at Nolop.

SAVE THE DATE

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Demo Day: show what you built

Final presentations and live product demos, built by students with no prior experience. Open to the public. No registration required.

Thu, Dec 10 | 1:30 pm | DEC

Week 13 · the finish line for your final project.



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Demo Day!

See the final presentations & live product demos built by students with no prior experience. Open to the public.



What your build must include

Your concept proposal has to show how you'll hit every one of these.

01

Laser cutting

At least one part designed as a 2D cut, and the proposal says where it fits in the build.

02

3D printing

Printed parts where they make sense: mounts, brackets, and custom shapes.

03

Microcontroller + I/O

A microcontroller with sensors and actuators, so the product senses the world and responds.

04

Smart API integration

AI or another external service wired into the device. A core maker skill.

THE BRIEF

Solves a unique problem, and fits in the project box: **40 × 25 × 25 cm**. Your proposal should clearly indicate which parts use each method and how they contribute to the design.

** Cannot be an automatic plant-watering system :)

Note-and-Vote: silent ideas, fast decision

01 · NOTE

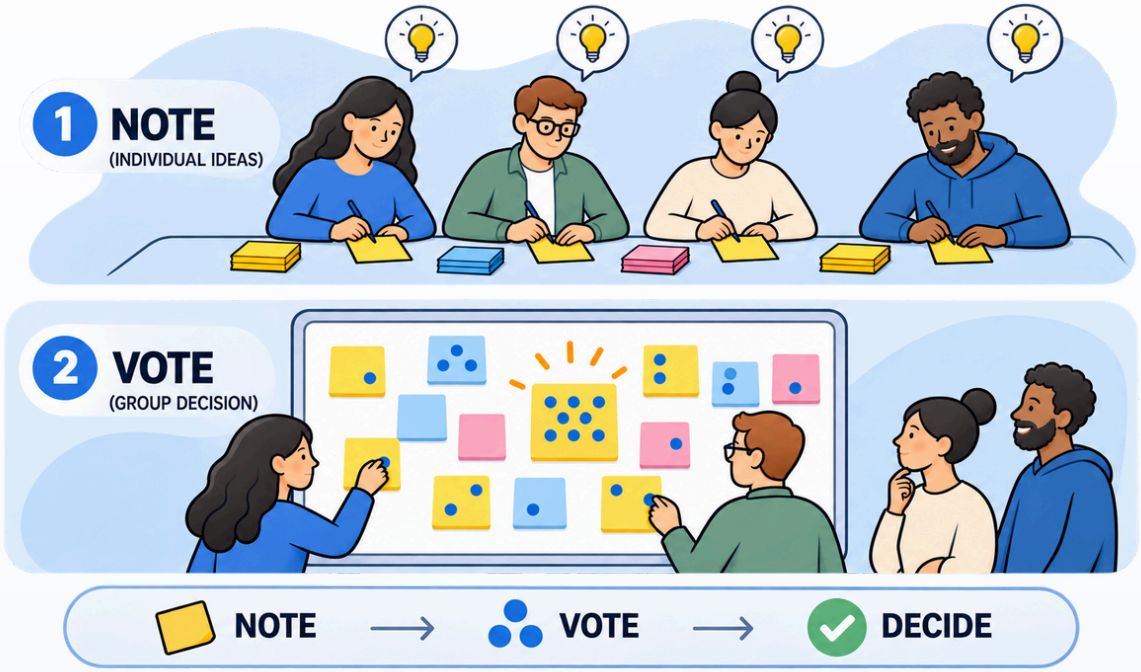
Ideas first, alone

Everyone writes their own ideas quietly, one per note, so the loudest voice doesn't set the agenda.

02 · VOTE

Vote with dots

Post the notes, then vote in silence with round stickers. The heatmap shows where the team's energy really is.



Voice for everyone, fast decision, then the team discusses the top ideas together.

From sticky notes to a team concept

STEP 1

Individually · 15 min

Write ideas

- → Write 3 to 5 unique ideas
- → One idea per sticky note
- → Each idea must meet the criteria

STEP 2

As a team · 15 min

Post & vote

- → Post every idea on the team wall
- → Briefly present your ideas to the team
- → Vote for your top 3 with round stickers. You can vote for your own

STEP 3

As a team · 15 min

Discuss & decide

- → Count the votes
- → Discuss the top 3 ideas
- → Combine ideas (optional)
- → Decide what you'll pursue

STEP 4

As a team · 45 min

Present

- → Present your top 1 or 2 ideas to the entire class
- → Get feedback from the instructors

What's due next class: Product Concept Presentation

01

Five minutes, in class

Every team pitches its product concept on Thu, Nov 12. Five minutes per team.

02

Show every requirement

How the concept uses laser cutting, 3D printing, a microcontroller with sensors and actuators, and smart API integration. It has to fit the 40 × 25 × 25 cm project box.

03

Product mockup

Bring sketches and a foam-core or other mockup of the concept.

04

Electronics and parts

List the electronics you'll need (sensors, motors, display, etc.) and what you'll laser cut and 3D print.

Part 1 of the final team project · 5 of the 35 team-assignment points. Slides due on Canvas by noon, Thu, Nov 12.

BEFORE YOU GO

Questions?

Next class: concept presentations, Thu, Nov 12.

WHERE TO FIND US

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Email
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Course materials
tuftsmaker.github.io/ENT-164